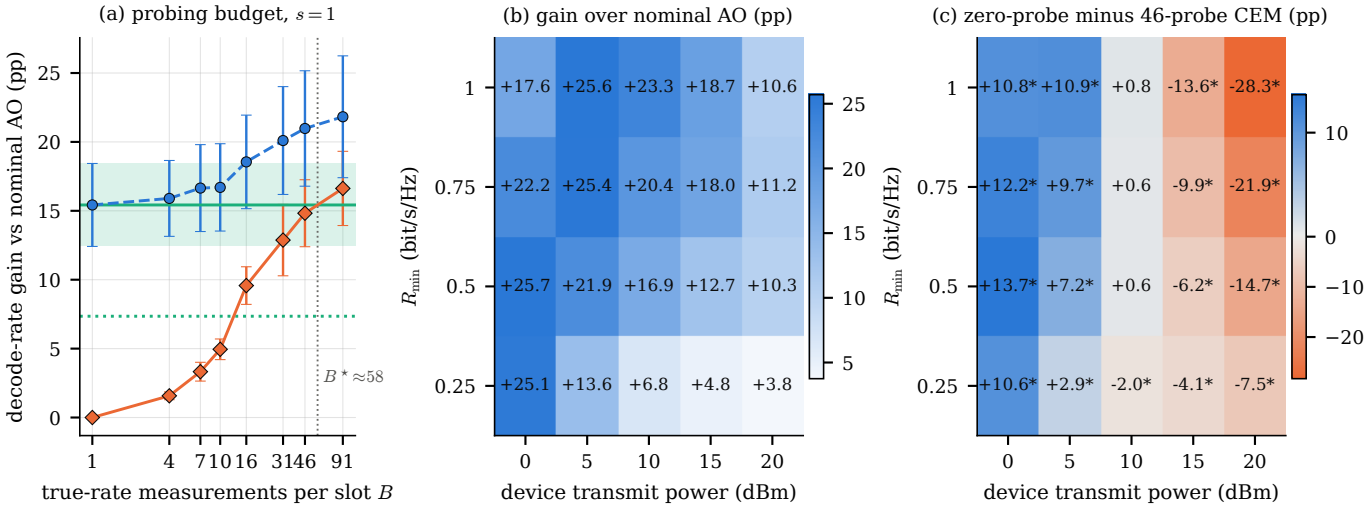
(a) probing budget, $s = 1$
decode-rate gain vs nominal AO (pp)
25
20
15
10
5
0
$B^* \approx 58$
1
4
7
10
16
31
46
91
true-rate measurements per slot B
(b) gain over nominal AO (pp)
$R_{\min}$ (bit/s/Hz)
1
0.75
0.5
0.25
+17.6 +25.6 +23.3 +18.7 +10.6
+22.2 +25.4 +20.4 +18.0 +11.2
+25.7 +21.9 +16.9 +12.7 +10.3
+25.1 +13.6 +6.8 +4.8 +3.8
25
20
15
10
5
0
5
10
15
20
device transmit power (dBm)
(c) zero-probe minus 46-probe CEM (pp)
$R_{\min}$ (bit/s/Hz)
1
0.75
0.5
0.25
+10.8* +10.9* +0.8 -13.6* -28.3*
+12.2* +9.7* +0.6 -9.9* -21.9*
+13.7* +7.2* +0.6 -6.2* -14.7*
+10.6* +2.9* -2.0* -4.1* -7.5*
10
0
−10
−20
0
5
10
15
20
device transmit power (dBm)
decode-aware AO, tuned width (no probes)
decode-aware AO, width 0.2 (no probes)
AO+CEM from max-min AO
AO+CEM from decode-aware AO